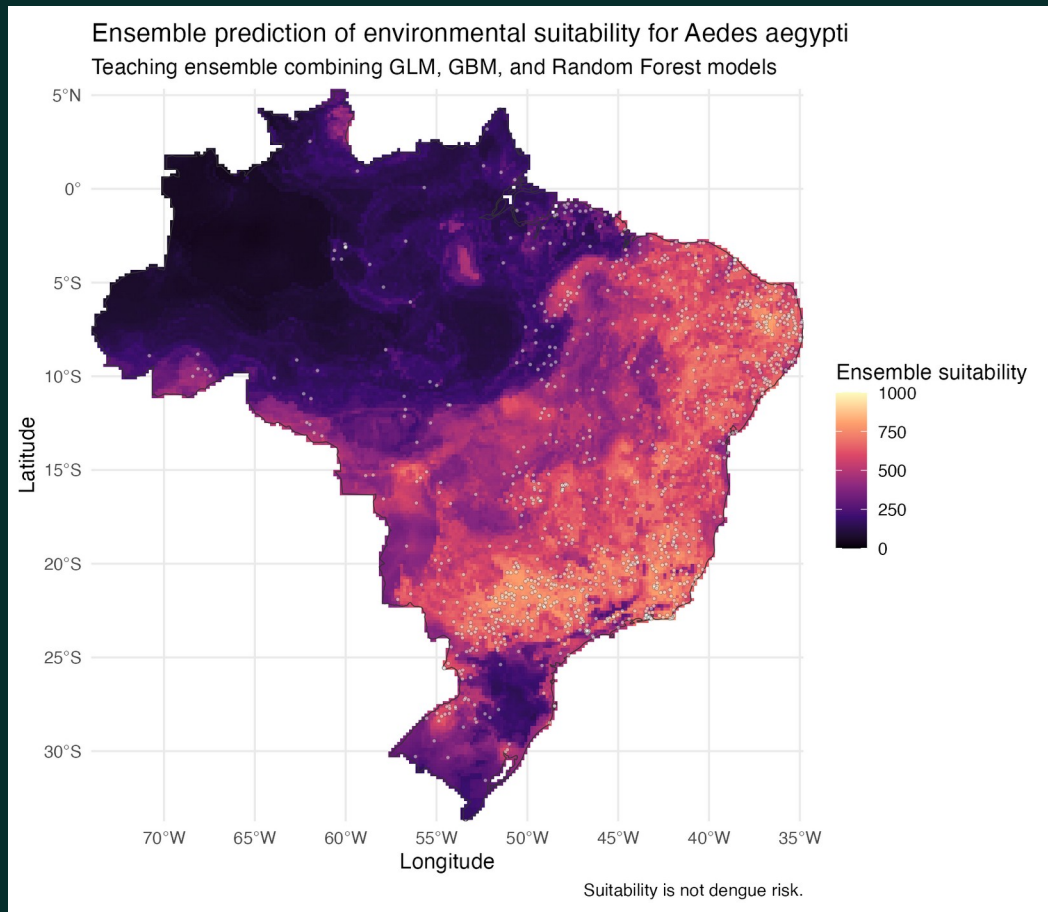


Aedes aegypti suitability modelling

A compact R workshop for infectious disease researchers: from GBIF records to biomod2 ensembles.

Brazil | GBIF | WorldClim | GLM, GBM, RF



The model answers a narrower question than dengue risk

For the workshop, we estimate where climate conditions are consistent with recorded *Aedes aegypti* occurrence.

Vector suitability

Environmental conditions that could support the mosquito.

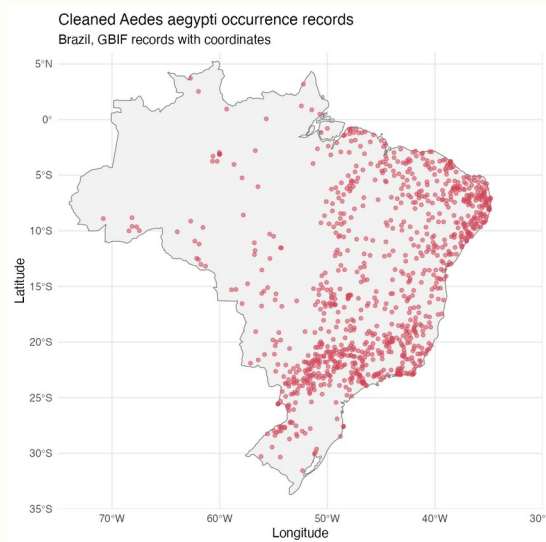


Dengue risk

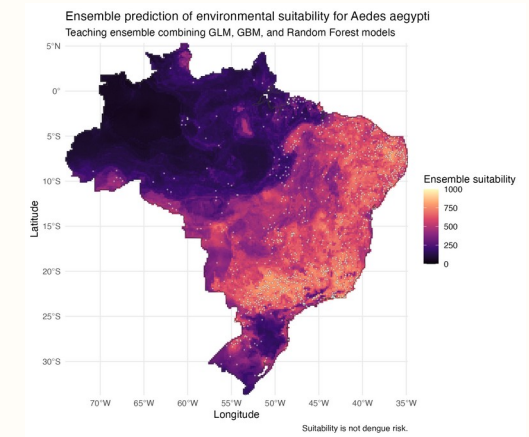
Also depends on people, virus circulation, immunity, housing, surveillance, and interventions.

SDMs translate a geographic pattern into environmental space

The algorithm does not know the biology. It sees locations, predictor values, and a modelling design.



geographic observations

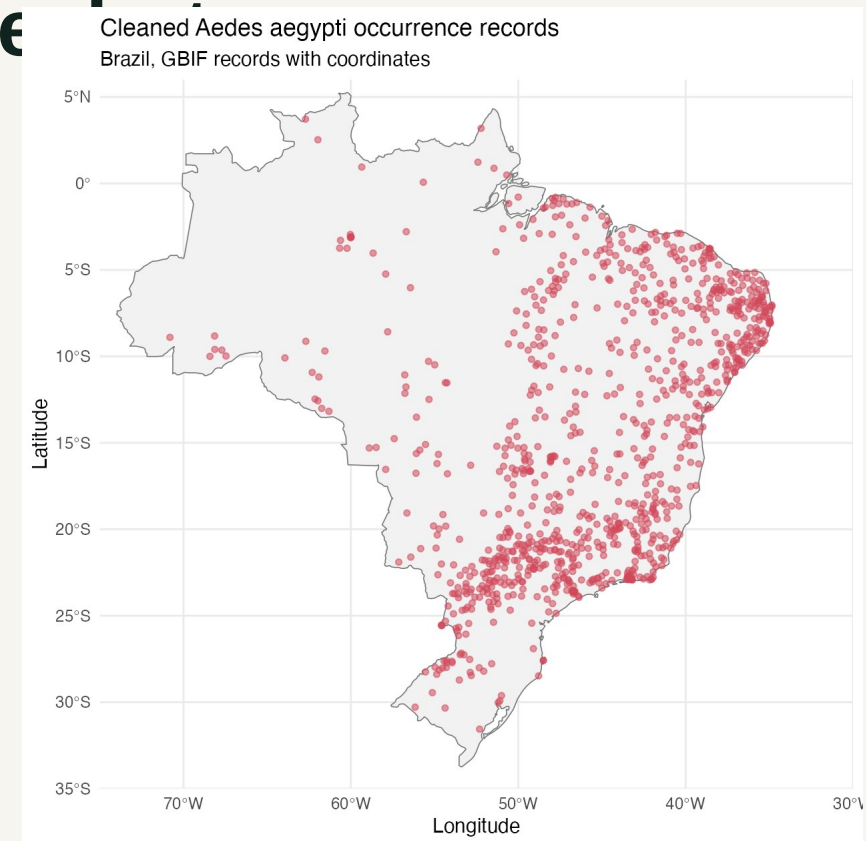


projected suitability

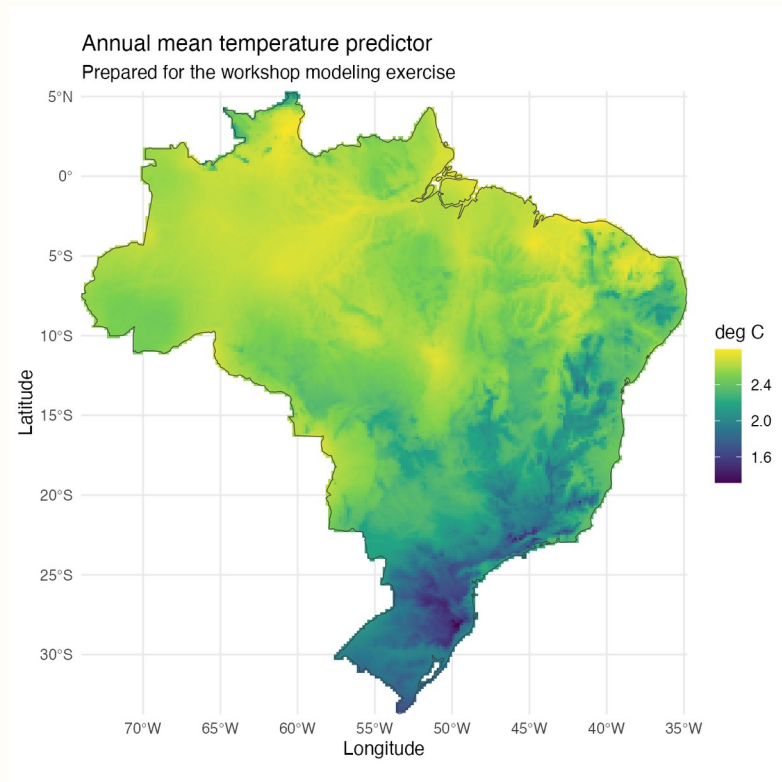
Occurrence records are evidence not a sampling design

GBIF gives us reach and reuse. It does not give us equal effort, perfect coordinates, or true absences.

- Remove missing and impossible coordinates.
- Thin duplicate records inside raster cells.
- Keep a cached clean file so the workshop runs.
- Interpret dense points as sampling effort and occurrence.



Predictors define the environmental lens



We deliberately use four predictors: enough to demonstrate the workflow, few enough to understand the model behaviour.

annual mean temperature
temperature seasonality
annual precipitation
precipitation seasonality

Teaching point: fewer variables make the first model easier to inspect.

Presence-only data force a modelling choice

Pseudo-absences are comparison points. They are not confirmed places where *Aedes aegypti* is absent.

**presence
records**
known points



pseudo-absences
background contrast



**biomod2 formatted
data**
ready for algorithms

The three example algorithms bring different biases

The goal is not to crown a winner. It is to see how modelling assumptions affect predictions.

GLM

**transparent
statistical trend**

simpler shape

GBM

**boosted sequence
of trees**

flexible shape

RF

**many trees voting
together**

robust shape

Evaluation is a stress test, not a certificate

Training data

fitted model

test data

TSS

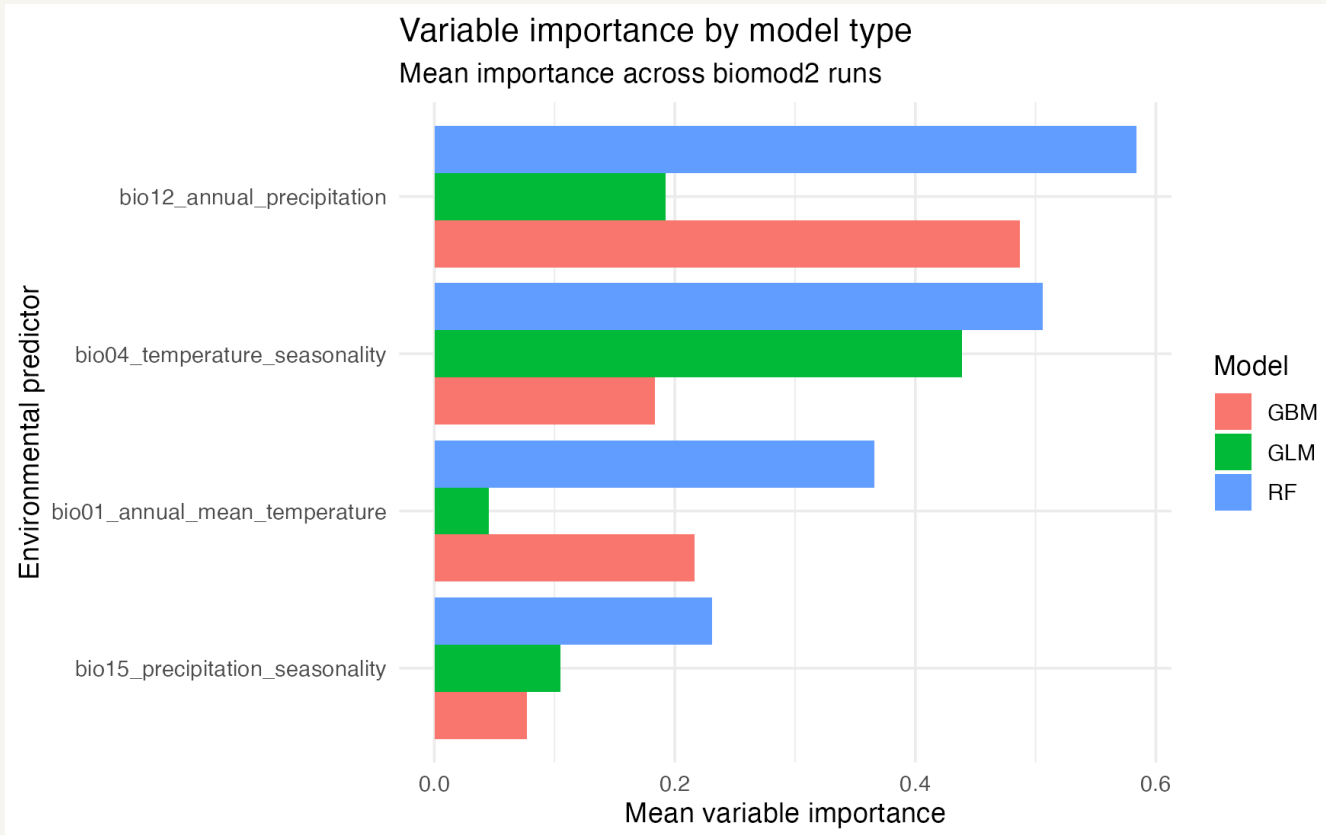
threshold-based balance
of sensitivity and
specificity

AUCro

threshold-independent
ranking of presences above
absences

Good scores make a model worth inspecting. They do not make the map true.

Variable importance turns outputs



Read as sensitivity

- Which predictors changed predictions most?
- Do GLM, GBM, and RF agree?
- Could this be ecology, sampling, or correlation among predictors?

The ensemble is a disciplined summary of disagreement

A mean ensemble reduces the map set; a weighted mean lets better-scoring models contribute more.

GLM

GBM

RF

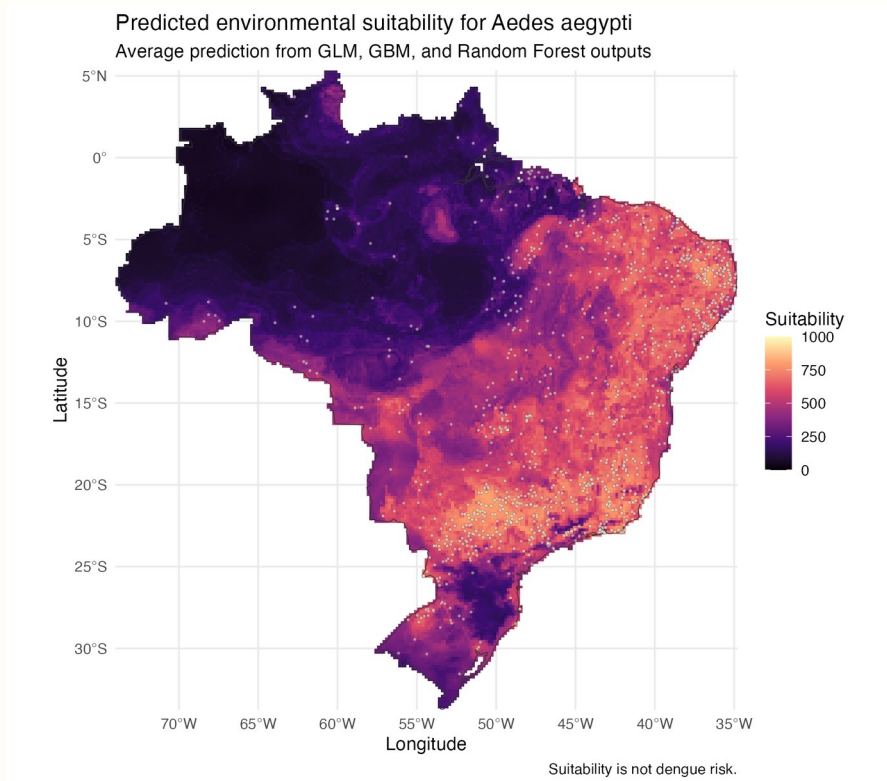
EMmean

average prediction

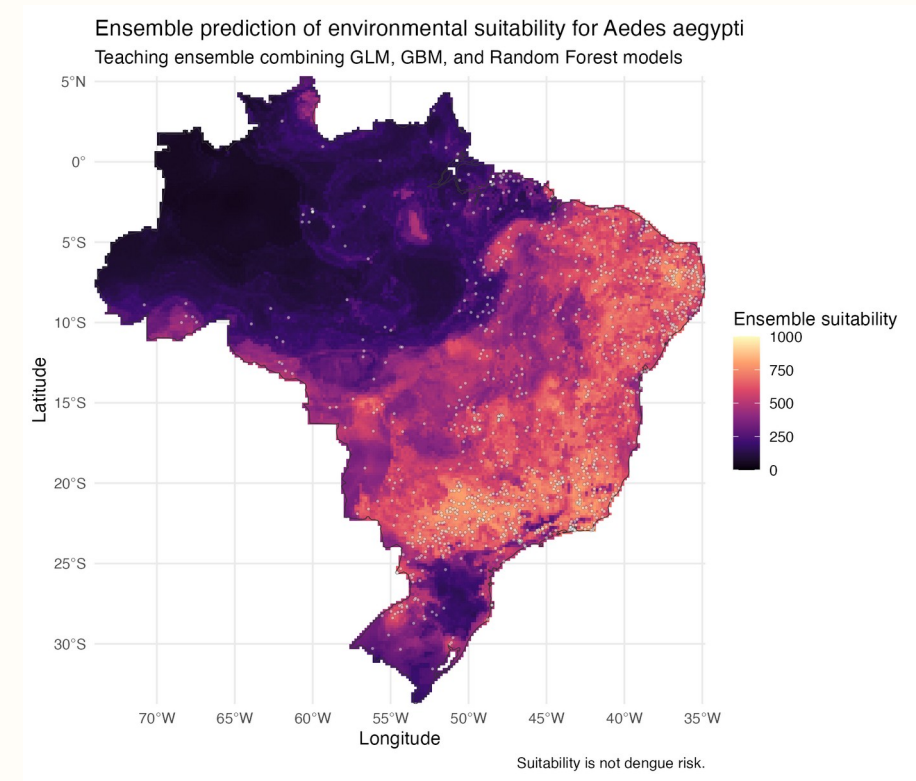
EMwmean

performance-weighted
prediction

Projection makes model choices visible

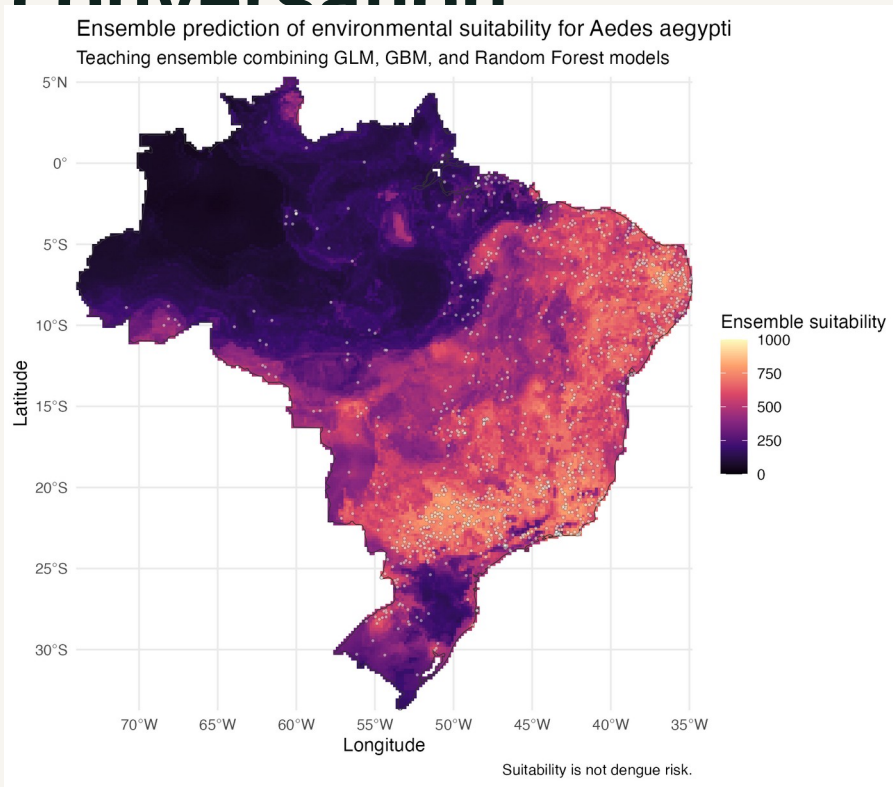


individual model average



ensemble projection

A good SDM should sharpen the next conversation



**Before acting on a map,
ask:**

- Where is suitability high or low?
- Where are the data thin?
- Where might sampling bias dominate?
- Which predictor story is plausible?
- What dengue surveillance data would you add?

**Now we build it line by line
in R.**